

1. Write a C program to find the factorial value of any number using recursion.

```
#include <stdio.h>

// Function Prototype
int fact(int);
int main(void)
{
    int n, f;
    printf("Enter a number: ");
    scanf("%d", &n);
    // Function call
    f = fact(n);
    printf("%d! = %d\n", n, f);
    return 0;
```

```
}
```

```
// Recursive function definition
```

```
int fact(int n)
```

```
{
```

```
if (n == 1)
```

```
{
```

```
    return 1;
```

```
}
```

```
else
```

```
{
```

```
    // Recursive step:  $n * (n-1)!$ 
```

```
    return (n * fact(n - 1));
```

```
}
```

```
}
```